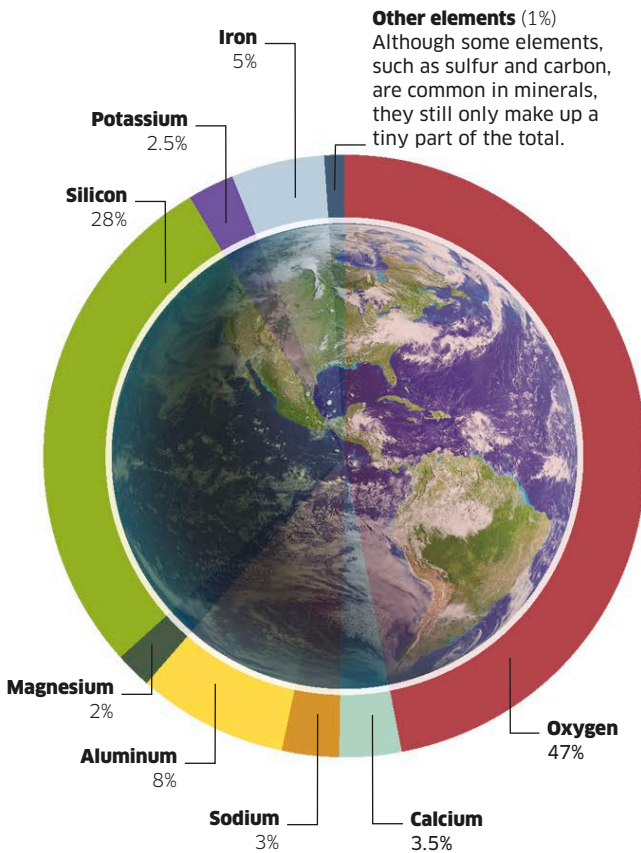


Elements of Earth's crust

Planet Earth is mostly made up of the elements iron, oxygen, silicon, and magnesium, with most of the iron concentrated in Earth's core. But Earth's outer layer, the crust, is made from minerals of many different elements, such as silicates (containing silicon and oxygen). This diagram shows which elements are most common in the crust.



Native elements

In Earth's crust, most elements exist combined with others in mineral compounds. But some, called native elements, appear in pure form. About 20 elements can be found in pure form, including metals, such as copper and gold, and non-metals, such as sulfur and carbon.



Sulfur
Powder and crystals of pure sulfur from volcanic gases accumulate around volcanic vents. In the rock cycle, it gets mixed into rocks. It also forms part of many mineral compounds.

Mineral compounds

There are more than 4,000 different kinds of minerals. Scientists classify them according to which elements they contain, and sort them into a few main groups. The group name tells which is the main element in all minerals in that group. All sulfide minerals, for example, contain sulfur. Many minerals exist in ores—rocks from which metals can be extracted—or as pretty gem crystals (see p.24).

Hematite

This oxide contains lots of iron, making it an important iron ore.



Oxides

Different metals combine with oxygen to form these hard minerals. They are in many ores, making these valuable sources of metal. Many make fine gems.

Baryte

The element barium combined with sulphur and oxygen makes baryte, which comes in many different forms.



Sulfates

A sulfur and oxygen compound combines with other elements to form sulfates. Most common are gypsum, which forms cave crystals (see pp.26–27), and baryte.

Malachite

Copper combines with carbon and oxygen to give this useful and decorative mineral its green color.



Carbonates

Compounds of carbon and oxygen combine with other elements to form carbonates. Many are quite soft. Some exist in rocks such as chalk and limestone.

Rose quartz

This is a pink form of quartz, one of the silicates made up of only silicon and oxygen.



Silicates

All silicates, the most common group, contain silicon and oxygen. Some include other elements, too. The rock granite is made of three silicates, including quartz.

Chalcopyrite

Both copper and iron can be sourced from ores containing this sulfide.



Sulfides

Metals combined with sulfur, but no oxygen, form sulfides. Sulfides make up many metal ores. Many are colorful, but are usually too soft to use as gemstones.

Fluorite

Calcium and fluorine make up this mineral, which comes in many different colors.



Halides

These minerals contain one or more metals combined with a halogen element (fluorine, chlorine, bromine, or iodine; see p.40). Rock salt is an edible halide.